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DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY - IST

Assignment Report on

Creation and Distribution of a Torrent File

for

DISTRIBUTED SYSTEMS AND COMPUTING – IT23601

Submitted by

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Creation and Distribution of a Torrent File in a Peer-to-Peer Environment

1. Objective

The objective of this assignment is to study and implement peer-to-peer (P2P) file sharing using BitTorrent. A torrent file is created, seeded by one peer, downloaded by another peer, and verified for data integrity.

2. Introduction

Peer-to-peer (P2P) systems allow computers (peers) to share resources directly without a central server. BitTorrent is a popular P2P protocol that divides files into small pieces and distributes them among peers.

This approach improves download speed, reduces server load, and increases fault tolerance. In this assignment, a torrent-based file-sharing environment is created using qBittorrent to demonstrate real-time P2P communication.

3. Tools Used

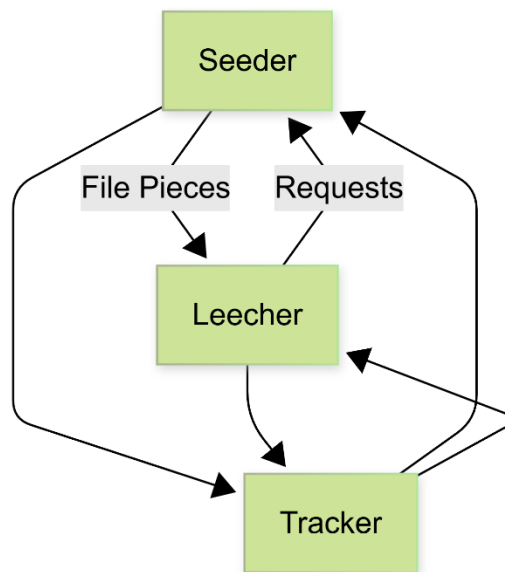
- Torrent Client: qBittorrent
- Operating Systems: Windows (Seeder), Ubuntu (Downloader)
- File: Dummy PDF (50 MB)
- Tracker: Public Trackers
- Network Model: Peer-to-Peer

4. System Architecture

The system follows a decentralized architecture where peers communicate directly with each other.

- One peer acts as **Seeder**
- One peer acts as **Leecher**
- A tracker helps peers find each other
- File is transferred in pieces
- Hashing ensures integrity

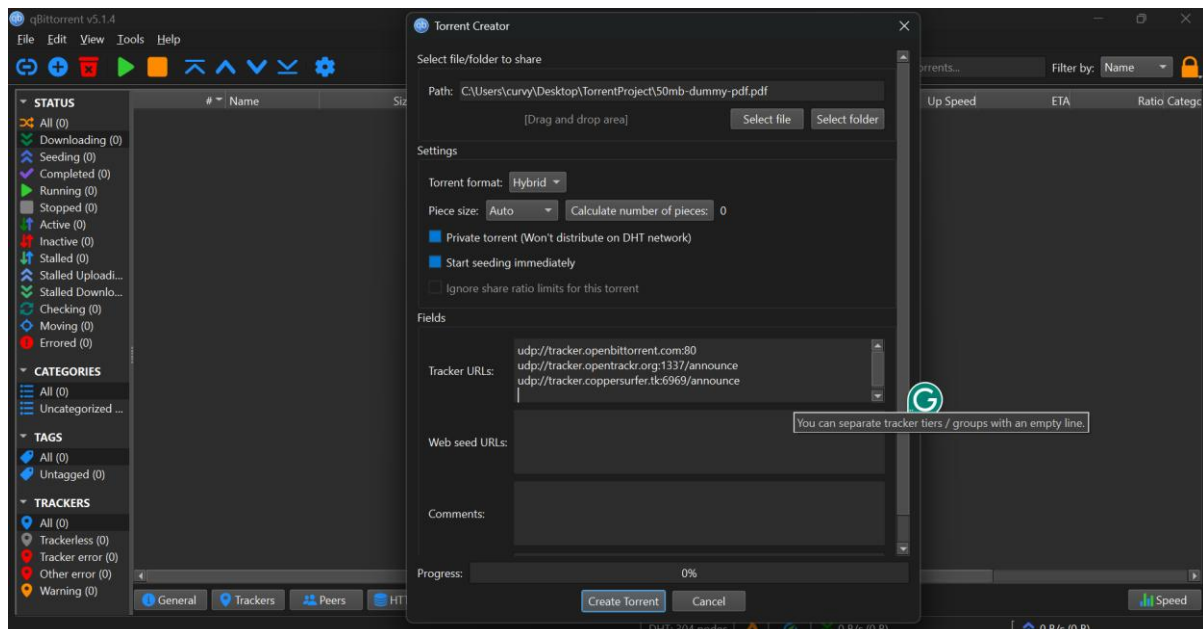
P2P Torrent Architecture



5. Implementation Steps

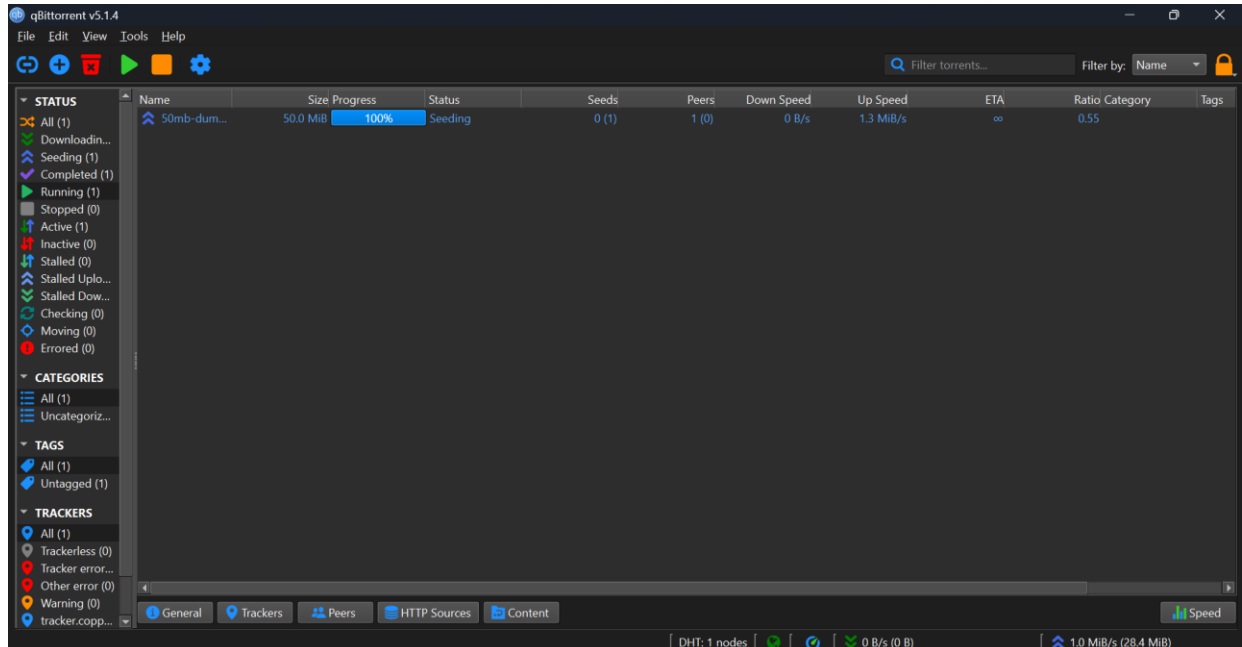
5.1 Torrent File Creation

The torrent file was created using the Torrent Creator tool in qBittorrent. A 50 MB dummy PDF file was selected, and public trackers were added. The .torrent file contains metadata about the file, such as file name, size, and hash values.



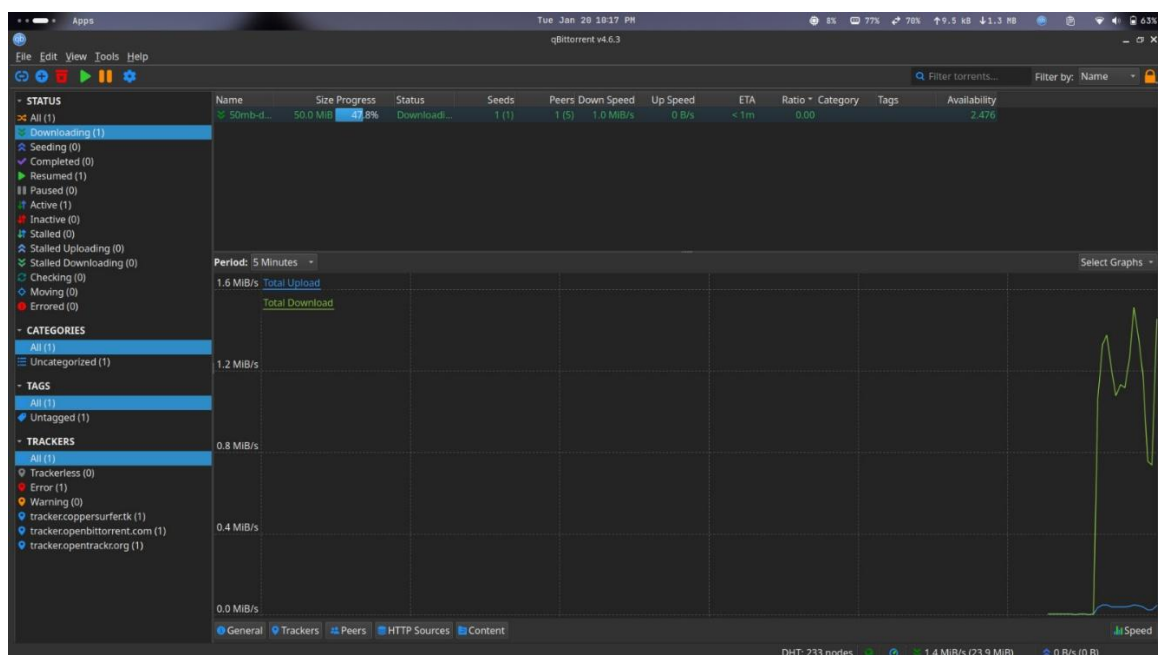
5.2 Seeding Process

After creating the torrent file, it was opened in qBittorrent. The location of the original PDF file was selected, and the system started acting as a seeder. The seeder makes the file available to other peers.



5.3 Downloading on Another Peer

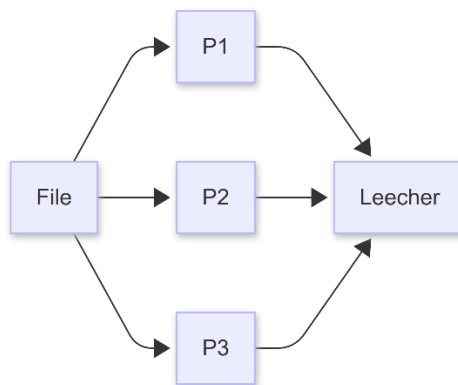
The same torrent file was opened on another system running Ubuntu. The downloader connected to the seeder through the tracker and started receiving the file in small pieces. These pieces were reassembled to form the complete file.



6. Peer-to-Peer Characteristics Observed

- No central server involvement
- Direct communication between peers
- File is transferred in small pieces
- Multiple peers can share data
- Hash-based data verification
- Decentralized resource sharing

File Piece Transfer



7. Relation to Distributed Systems

Concept	Explanation
Decentralization	No single controlling node
Scalability	Performance improves with more peers
Fault Tolerance	System continues even if one peer fails
Data Integrity	Hash values ensure correctness

8. Advantages of Torrent-Based File Sharing

- Faster downloads due to parallel piece transfer
- No single point of failure
- Efficient bandwidth utilization
- Built-in error detection
- Suitable for large file distribution

9. Conclusion

This assignment successfully demonstrated peer-to-peer file sharing using BitTorrent. A torrent file was created, seeded, downloaded, and verified without using a centralized server. The experiment provided a clear understanding of how distributed systems function in real-world applications.
